EW1: Final Evaluation

GOOGLE DINOSAUR GAME

- **Team members names:**
  Ganipisetty Nischal &
  Batta Venkata Shashank
- **Team ID:** 39
- **TA:** Vedant sir

Motivation or Background

- Engaging learning tool
- Hands-on experience with systems
- Hardware-software integration
- Resourcefulness
- Skill building
- Universal appeal

ENGAGING LEARNING TOOL

The project offers an interactive way to learn microcontroller programming, circuit design and game development

HANDS-ON EXPERINCE WITH SYSTEMS

The project provides a good interaction with components like LCD Display, push button, buzzer, I2C

HARDWARE-SOFTWARE INTEGRATION

Translating a software-based game into a physical, hardware-driven system

RESOUECEFULNESS

Teaches the art of maximising functonality within the constraints of arduino's limited processing power and memory

SKILL BUILDING

Equips developers with skills in coding, electronics, and problem-solving, applicable to more complex embedded systems.

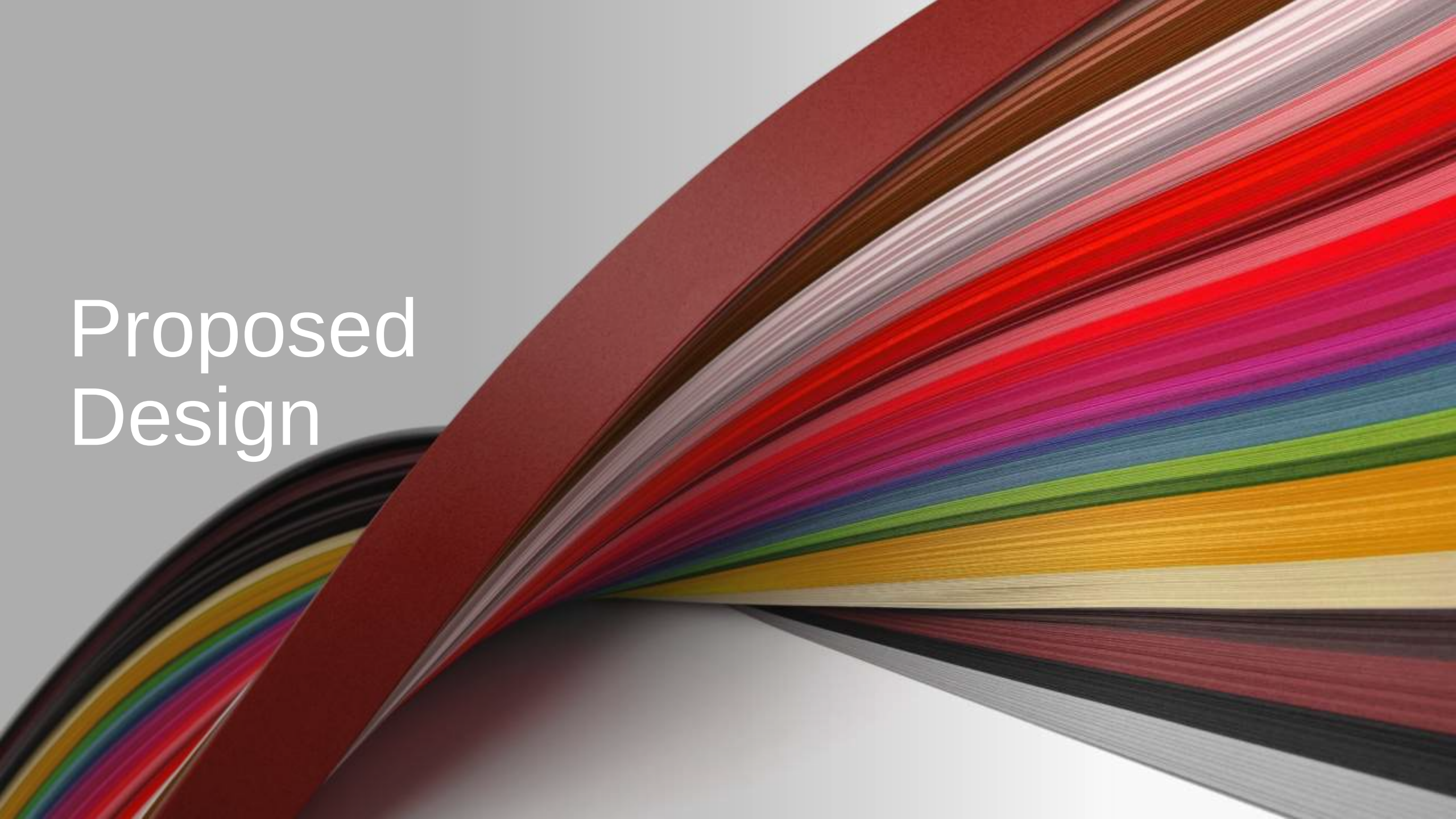
UNIVERSAL APPEAL

The game is universally recognizable, making it relatable and enjoyable for a broad audience during demonstrations.

Problem Statement

- Create a dinosaur and random obstacles
- Real-time interaction with game
- Displaying score at end of the game

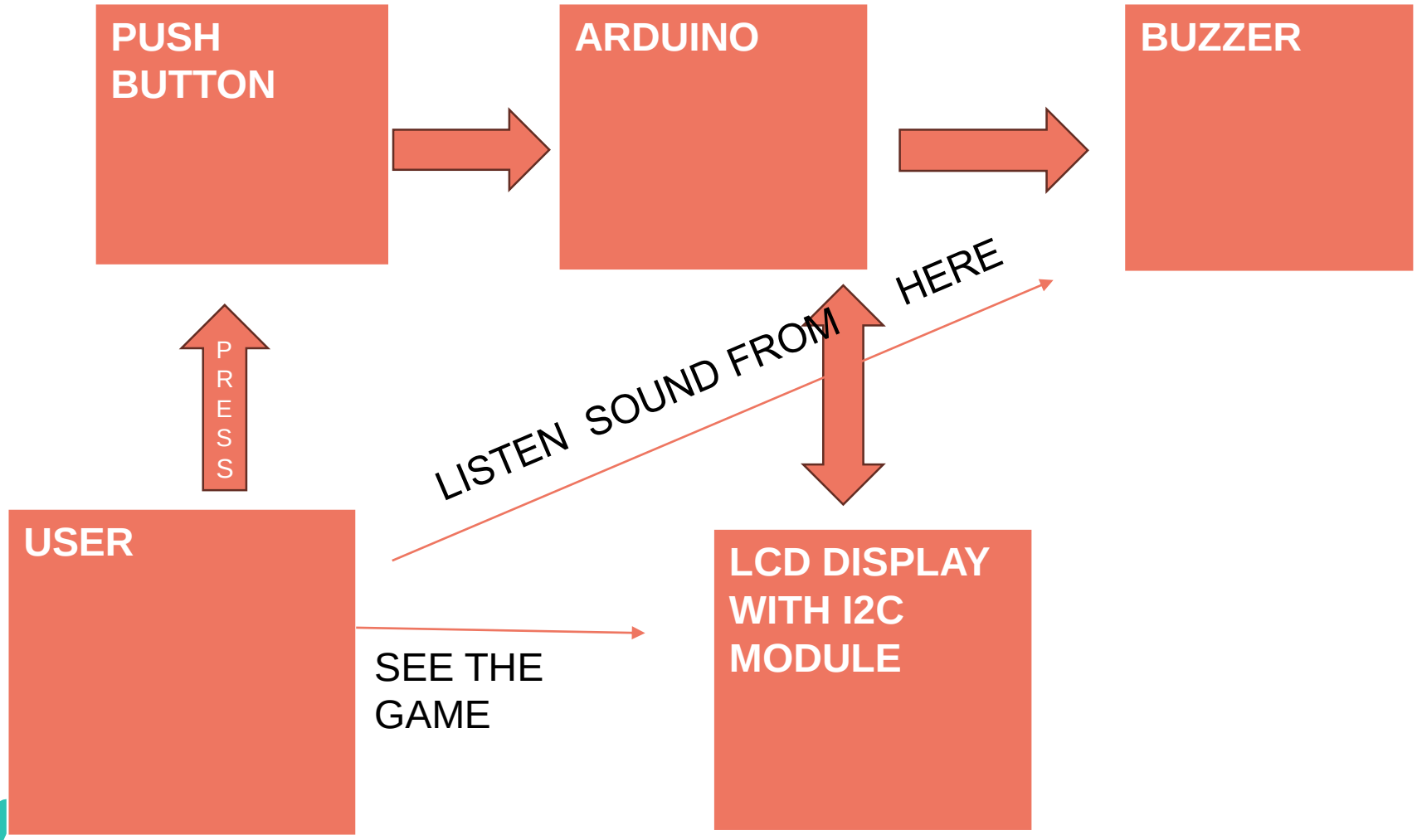
Proposed Design



Mapping of requirements

- To create a dinosaur and obstacles we need an "*Arduino*"
- To avoid the collision between them we need a "push button"
- And we need a "buzzer" to give signals that game starts or game overs or either it crosses the obstacle or not
- We need an "LCD" Screen to display score and all the game

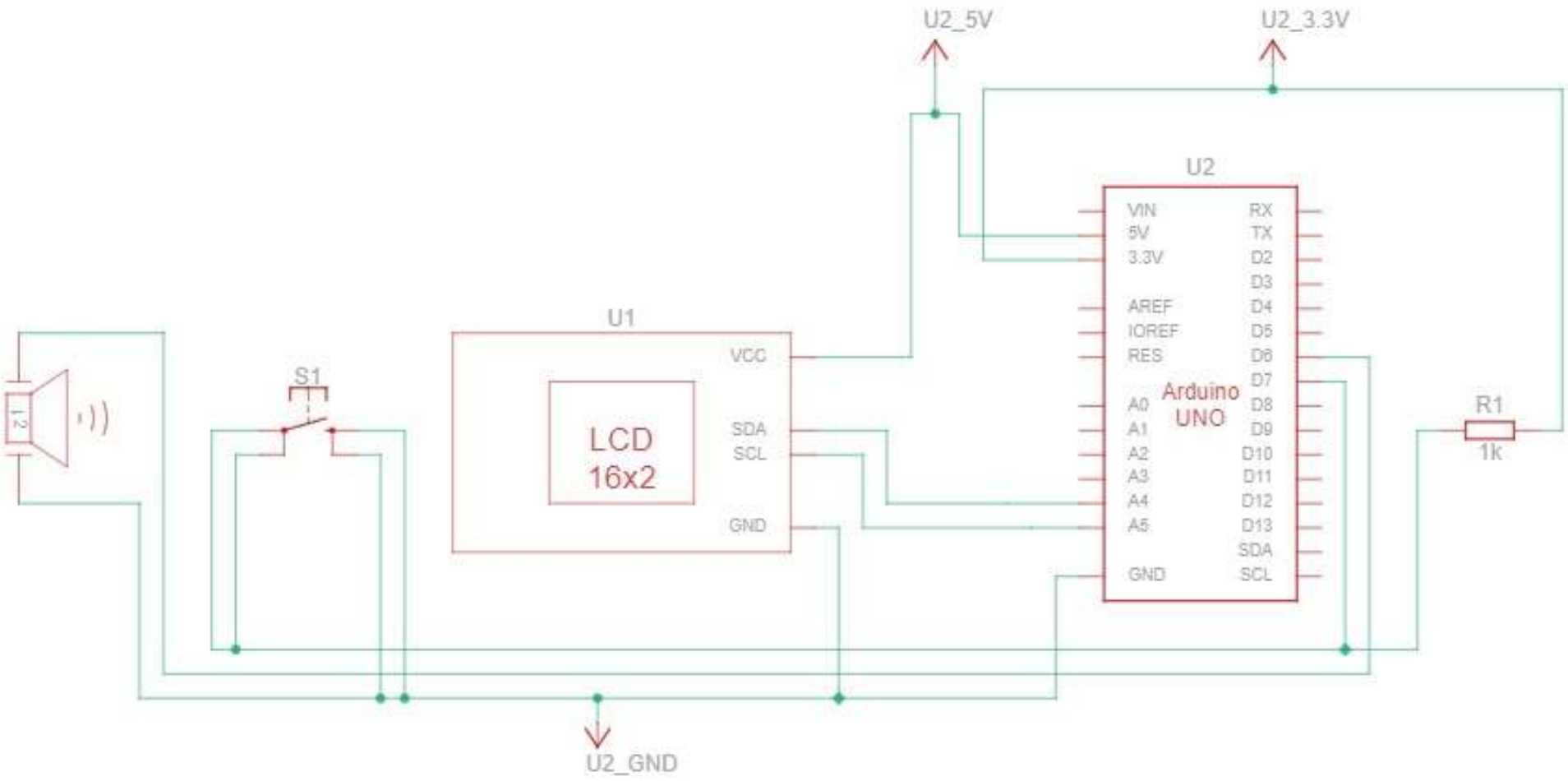
Block diagram



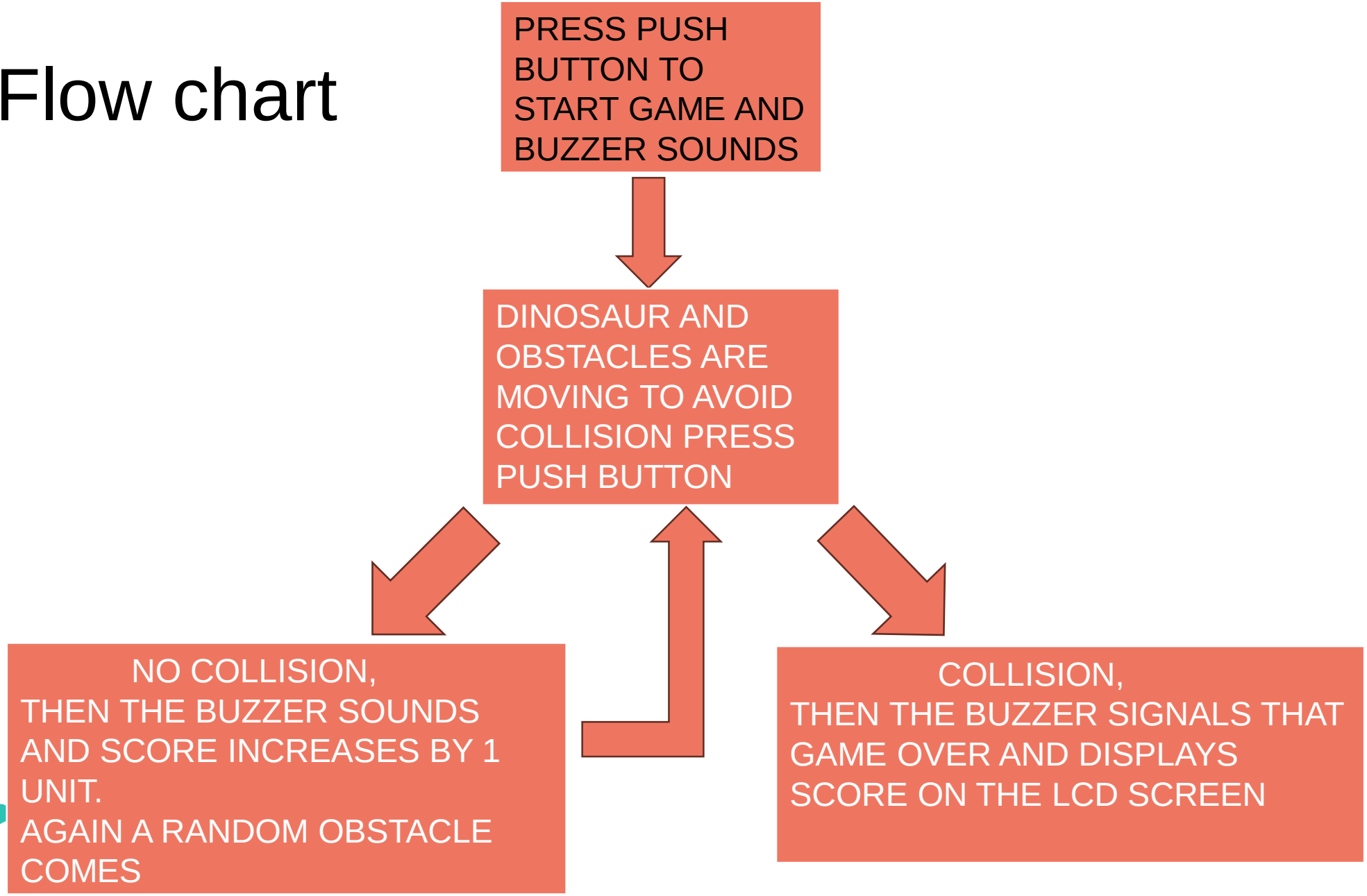
Explanation

- The game displays on LCD Screen and it is seen by the user
- The user press the push button to start the game
- The user press the push button at the time of colliding dino and obstacles
- The user listen the sound when dino crosses obstacle

Circuit diagram

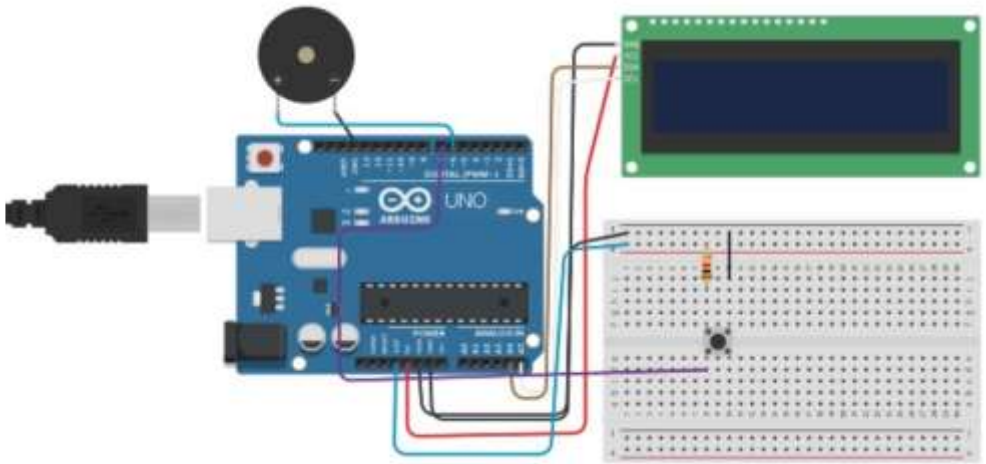


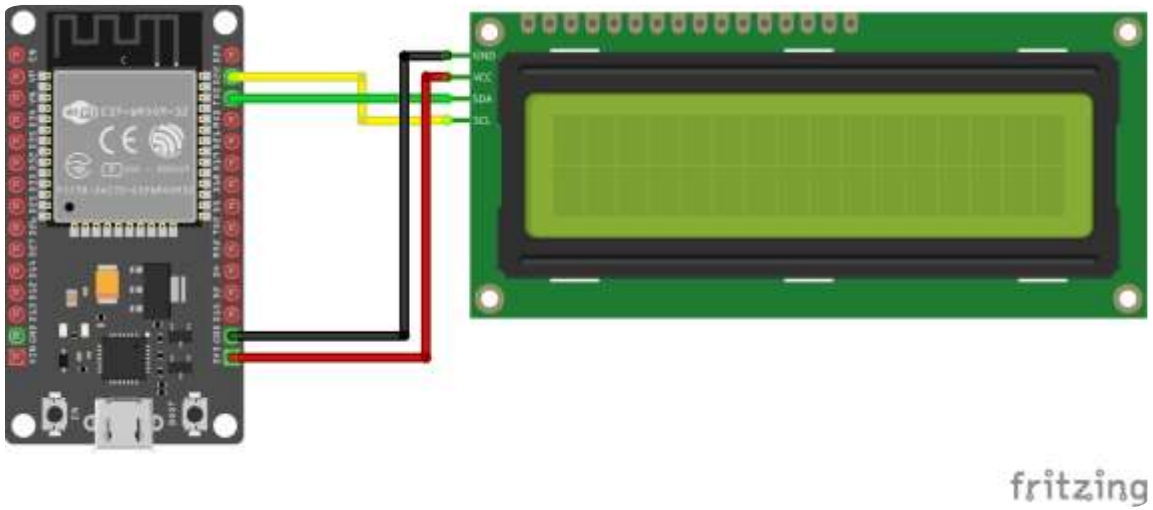
Flow chart


PRESS PUSH BUTTON TO START GAME AND BUZZER SOUNDS
DINOSAUR AND OBSTACLES ARE MOVING TO AVOID COLLISION PRESS PUSH BUTTON
NO COLLISION, THEN THE BUZZER SOUNDS AND SCORE INCREASES BY 1 UNIT. AGAIN A RANDOM OBSTACLE COMES
COLLISION, THEN THE BUZZER SIGNALS THAT GAME OVER AND DISPLAYS SCORE ON THE LCD SCREEN

Demo

- Video Link : https://photos.app.goo.gl/aKwjLF4nArZE4kzR9





Key performance indicators

- We used I2C Module to make easier connections rather than direct LCD connections.
- To easy access for users we used PCB and soldered the wires instead of Bread board

Cost of the solution

Arduino Uno

INR 600

Push button

INR 10

Buzzer

INR 99

Jumper Wires

INR 100

PCB

INR 11

Total Cost: INR 820

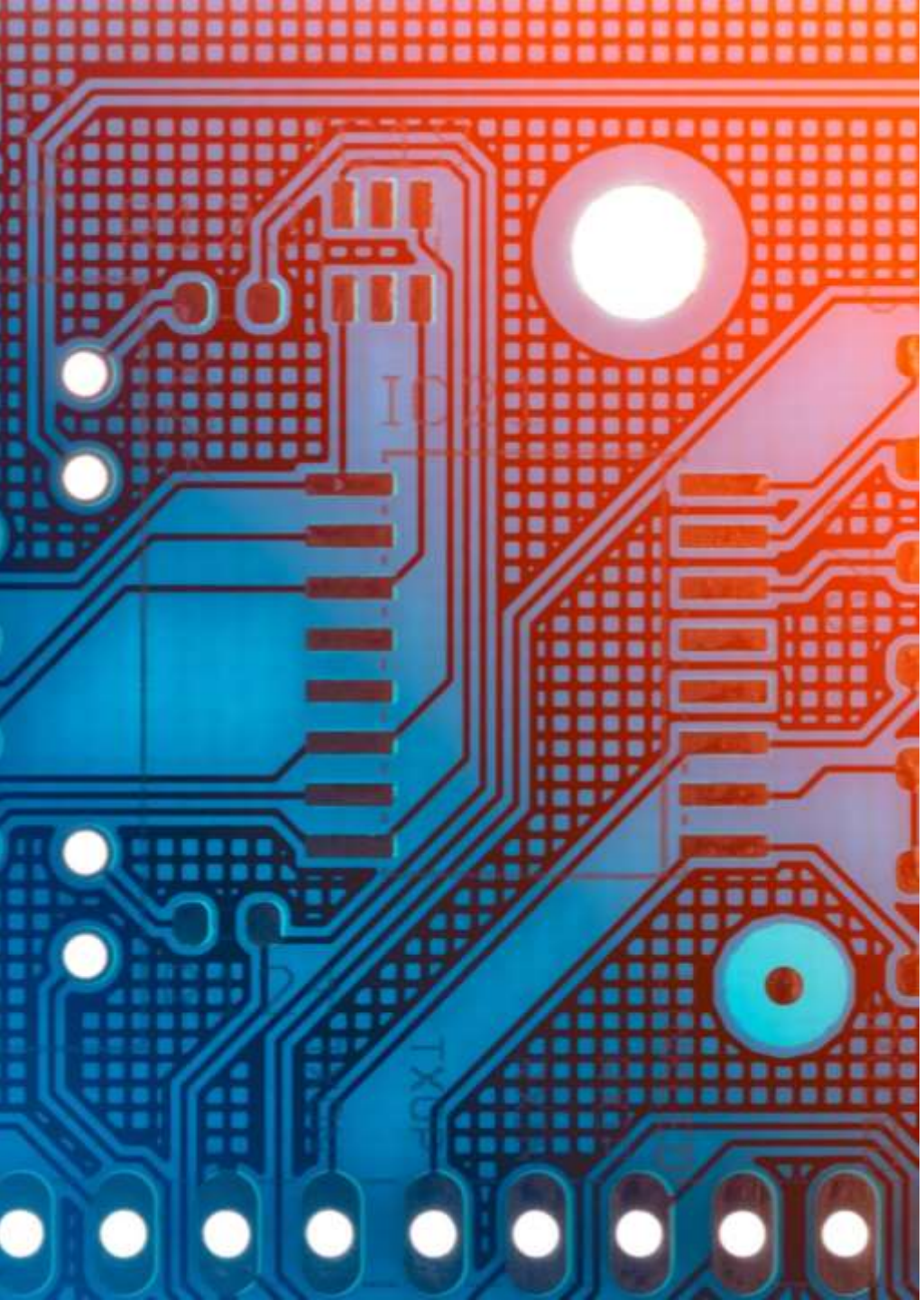
Contribution from team members

- **Member1:** Software Programming (Mostly)
- **Member2:** Presentation and reports (Mostly)
- **Joint effort:** Circuit Design and Hardware assembly

Remarks

Technical challenges:

- We firstly created dinosaurs and obstacles separately. We faced problem while combining both in one frame.
- We also faced problem for showing the high score.



Learning from the project

- Practical understanding of Arduino, I2C module, Push button, Buzzer
- Programming and Coding
- Hardware-software integration
- Importance of teamwork in project development

References

- Official documentation for Arduino and Liquid Crystal library(I2C Module)
- Online tutorials for I2C Module, LCD Display
- Open AI for debugging the code and for sound effects

Thank You...!